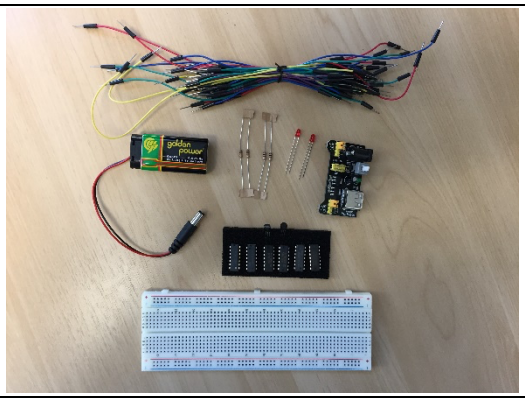
Lab 2

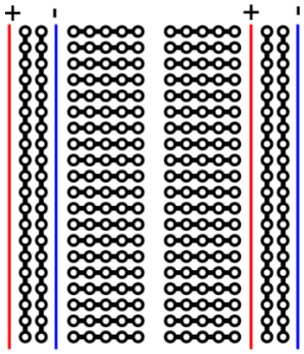
Objective: Build some simple circuits and use a transistor as a switch.
This lab is preliminary work for labs 3 and 4.

Each kit should contain:

2 * LED (light-emitting diode) 2 * 1K ohm resistor (Brown Black Red Gold) 2 * 10K ohm resistor (Brown Black Orange Gold) 1 * breadboard 1 * power board 1 * 9v battery (and connector) 2 * CTBC 547 JS Transistor (NPN) 2 * 74LS86 (quad 2-input XOR gate) 2 * 74LS08 (quad 2-input AND gate) 1 * 74LS00 (quad 2-input NAND gate) 1 * 74LS32 (quad 2-input OR gate) Bundle of wires	

The Breadboard

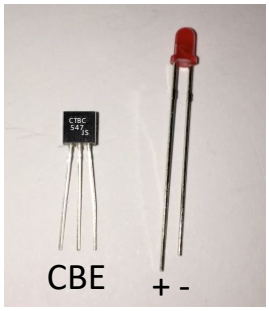
Breadboards are designed for experimenting with electronic circuits. They are a pre-printed set of connections inside a piece of moulded plastic designed to allow the plug-in of power, resistors, LEDs, chips, and so on. The breadboards we are using are connected like this:



From (http://www.backward-workshop.com/electronics/breadboard-curriculum/breadboard/)

LED and Transistor

The LED and Transistor are *polarised* so have a particular orientation. The LED is a diode, allowing electricity to flow in one direction only (from + to -). The Transistor allows flow from the collector (C) to the emitter (E) if current is applied to the base (B).



Units and Essential Formulae

Quantity:	Symbol:	Unit:
Voltage	V	volt (V)
Current	I	amp (A)
Resistance	R	ohm (Ω)
Power	P	watt (W)

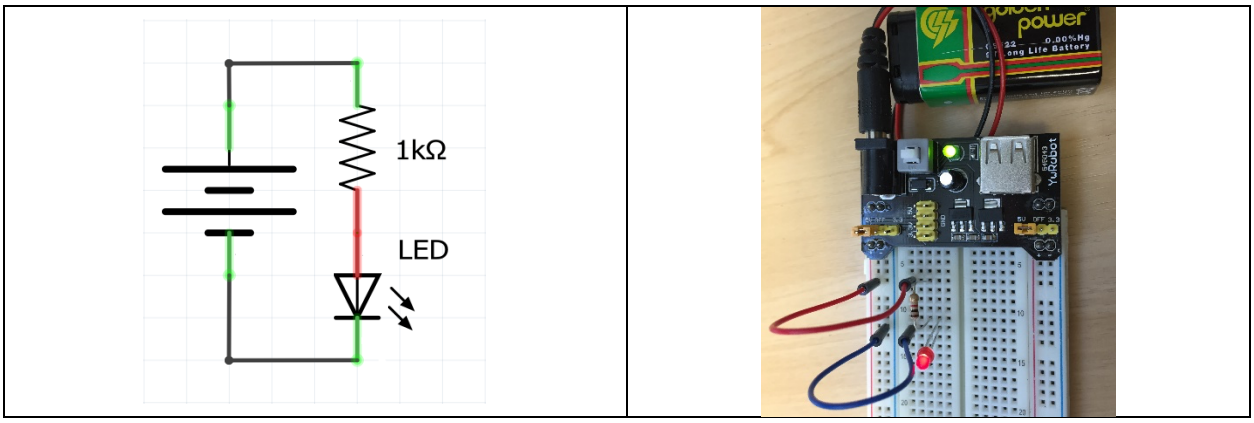
$V = I \times R$ (Ohm's Law)

$P = V \times I$

Exercise 1: An LED is a Light

In this exercise we'll get familiar with the breadboard, LED, wire, and powering it up. We're going to connect the LED to the power supply and turn it on. The circuit schematic is on the left; the finished physical circuit is on the right. You need to be aware that an LED is a diode and so only works one way in the circuit. The positive end of the LED is the long "leg".

The reason for the resistor is to limit the current so that the LED doesn't burn out. Be aware that the power supply board regulates (limits) the 9 V supplied by the battery to a more suitable 5 V.

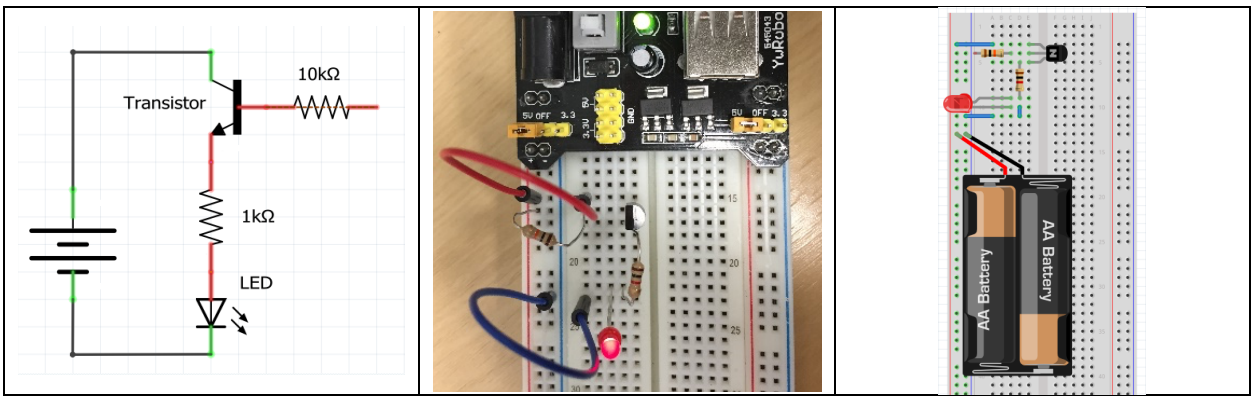


Next, try swapping the LED and the resistor (maintaining the LED's orientation), and see what difference it makes.

Exercise 2: A Transistor is a Switch

In this exercise we're going to use a transistor as a switch to turn on and off the LED. This is achieved by adding the transistor to the circuit between the battery and the resistor, then using the base (the middle pin) to apply power to the transistor to allow current to flow through it. The base resistor is used to protect the transistor.

Once you've built this circuit, use the base of the transistor (the middle pin) to turn on and off the LED. To do this, run a wire from the free end of the 10 kΩ resistor to either the + or – rail on the breadboard.



As before, try changing the order of the transistor, resistor and LED and see what difference it makes.

Lab Credit

Have a tutor/demonstrator check your work so you can get credit for this lab.

Summary

In this lab we first built a tiny torch, then we learned that the transistor is a switch. We used the middle pin of the transistor to turn the torch on and off. In the next lab we'll build a NAND gate and an AND gate using only transistors.